

Xiyao Cheng

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Xiyao Cheng

Google Scholar

Columbia, Missouri - 65203, US

EDUCATION

University of Missouri - Columbia, Mizzou

01/2021 - expected Dec 2026

Ph.D. in Computer Science

Columbia, USA

◦ Academic Advisor: Prof. Prasad Calyam Curators' Distinguished Professor, Center Director of AI Education, Research and Infrastructure (AERI)

◦ Dissertation: Knowledge Graph based Data Models for Scholar Recommendations in Academic Social Networks

◦ 2024 Outstanding Ph.D. Student in Computer Science Department

◦ GPA: 3.97/4.0

China Agricultural University, CAU

09/2016 - 01/2019

M.S. in Computer Science

Beijing, China

Tianjin Polytechnic University, TJPU

09/2012 - 06/2016

B.E. in Software Engineering

Tianjin, China

RESEARCH INTERESTS

- Computational Network Science
- Knowledge-driven Cyber Defense
- Graph & Hypergraph Learning for Complex Systems
- Human-Centered AI and Conversational Intelligence
- Knowledge Graphs for Security and Trust
- AI for Social Good
- Intelligent Distributed Systems

WORK EXPERIENCE

Center for Cyber Education, Research and Infrastructure (Mizzou CERI)

05/2021 – present

Research Assistant

Columbia, Missouri, USA

- Proposed and developed knowledge graph-based scholar recommendation systems leveraging deep learning and LLMs, with applications in cybersecurity, flexible manufacturing, and healthcare.
- Published 16+ peer-reviewed papers in top-tier venues including IEEE DSAA, IEEE TBD, ACM Computing Surveys, and CCPE.
- Contributed to NSF proposal writing on cybersecurity and flexible manufacturing in collaboration with faculty and postdoctoral researchers across multiple institutions.
- Mentored graduate and undergraduate students from diverse backgrounds in research methodology and project execution.
- Conducted sponsored research funded by NSF and Army Research Lab, collaborating with University of Southern California, Texas Tech University, and Missouri S&T.

MU Informatics and Data Science Institute

12/2025 – present

Research Assistant

Columbia, Missouri, USA

- Co-led development of an AI-driven scholar recommendation platform with Dr. Sounak Chakraborty (Fellow, International Statistical Institute) and Dr. Nilanjan Chakraborty, advancing methods for interdisciplinary research collaboration.
- Developed a hypergraph-based time-aware attention network and a relational scoring mechanism for team recommendation by jointly optimizing diversity, quality, and coverage and a deployed web application.

Artificial Intelligence and Complex Systems (AICS) group

01/2023 - 05/2023

Ph.D Student Research Aide

Columbia, Missouri, USA

- Collaborated with Dr. Mayank Kejriwal (Fellow, Information Science Institute) from University of South California to propose and develop a knowledge graph-based scholar recommendation system
- Constructed a heterogeneous knowledge graph from NSF award and Semantic Scholar publication data, and applied deep learning models to enable complex multi-hop scholar retrieval.
- Designed and conducted end-to-end experiments evaluating recommendation performance; benchmarked against multiple baselines to validate the effectiveness of the proposed approach.

Innovation Diffusion Lab

05/2022 - 05/2023

Ph.D Student Research Aide

Columbia, Missouri, USA

- Developed a domain-specific scholarly search platform for medical researchers and clinicians, including the KnowCOVID-19 Science Gateway and Vidura Chatbot.
- Designed and conducted end-to-end experiments demonstrating quantitative performance improvements over baselines.
- Performed a usability study with 15+ real users to evaluate system effectiveness in practical settings, collecting and analyzing survey-based feedback.

College of Information and Electrical Engineering

09/2016 - 01/2019

Research Assistant

Beijing, China

- Designed annotation guidelines and led annotators in labeling a Sina Weibo-sourced emotion-cause dataset, ensuring high inter-annotator agreement and data quality.
- Developed and evaluated machine learning models on the annotated emotion-cause corpus, demonstrating that the curated dataset enables high-performance emotion cause identification.
- Built and deployed a food safety information extraction application system combining rule-based and machine learning methods to extract structured story details from real-world news articles.

TEACHING EXPERIENCE

University of Missouri-Columbia

Jan - May 2022, Aug - Dec 2025

Teaching Assistant

- Assisted Prof. Prasad Calyam with teaching a course on *Cloud Computing*. Designed hands-on instruction labs that use public clouds such as e.g., AWS, GCP, as well as open clouds such as e.g., GENI, Chameleon, Nautilus, and CloudLab.
- Mentored and helped 80+ graduate and undergraduate students in class.
- Mean Course Evaluation Score: 4.7/5

University of Missouri-Columbia

Aug - Dec 2022, Jan - May 2023

Teaching Assistant

- Assisted Prof. Ries in teaching and grading a course on *Advanced Algorithm Design* (Fall 2022 and Spring 2023). Designed and taught lab lectures with 40+ undergraduate students.
- Responsible for developing, and managing course materials, holding office hours, and submitting and approving grades.

University of Missouri-Columbia

Aug - Dec 2023

Teaching Assistant

- Assisted Dr. Ufuktepe in teaching and grading a course on *Algorithm Design and Programming I*. Designed and taught lab lectures with 100+ undergraduate students.
- Responsible for developing, and managing course materials, holding office hours, and submitting and approving grades.

University of Missouri-Columbia

Aug - Dec 2023

Teaching Assistant

- Assisted Dr. Micheal Jurczyk in teaching and grading a course on *Computer Network*. Mentored 80+ undergraduate students and graduate students.
- Responsible for lecturing, developing, and managing course materials, holding office hours, and submitting and approving grades.

JOURNAL PAPERS

- [1] **Cheng, X.**, Rayhan, M., Pandey A., Zhang Y., Kejriwal M., & Calyam P., *ScholarInfluencer: AI-driven Role-Aware Scholar Recommendations based on Temporal Graph Networks*. ACM Transactions on Knowledge Discovery from Data, 2026. *Under Reviewing*. **Impact Factor: 4.9**
- [2] **Cheng, X.**, Rayhan, M., Edara Lakshmi S., Pandey A., Zhang Y., Kejriwal M., & Calyam P., *Knowledge Graph-based Modeling for Identifying Scholar Teams in Academic Social Networks*. IEEE Transactions on Artificial Intelligence, 2026. *Under Reviewing*. **Impact Factor: 6.4**
- [3] Zhang, Y., **Cheng, X.**, Oruche, R., Sivarathri, S. S., & Calyam, P. *ScholarFinder: Knowledge Embedding Based Recommendations Using a Deep Embedded Clustering Model*. IEEE Transactions on Big Data, 2025. **Impact Factor: 5.7**
- [4] Oruche, R., Goruganthu, S. K., Akula, R., **Cheng, X.**, Md Goni, A., Shibo, B. W., Kee, K., Zampieri, M., & Calyam, P. *A Survey on the Recent Advancements in Human-Centered Dialog Systems*. ACM Computing Surveys, 2025. **Impact Factor: 28.0**
- [5] Oruche, R., **Cheng, X.**, Zeng, Z., Vazzana, A., Goni, M. A., Shibo, B. W., Goruganthu, S. K., Kee, K., & Calyam, P. *Chatbot Dialog Design for Improved Human Performance in Domain Knowledge Discovery*. IEEE Transactions on Human-Machine Systems, 2025. **Impact Factor: 4.9**
- [6] Neupane, R. L., Calyam, P., Wang, S., Neupane, K., Pandey, A., **Cheng, X.**, Gafurov, D., et al. *Online Self-Service Learning Platform for Application-Inspired Cloud Development and Operations (DevOps) Curriculum*. IEEE Transactions on Learning Technologies, 2024. **Impact Factor: 4.9**
- [7] Oruche, R., Milman, E. D., Lemus Alarcon, M., **Cheng, X.**, Pandey, A., Wang, S., Calyam, P., & Kee, K. *Science gateway adoption using plug-in middleware for evidence-based healthcare data management*. Concurrency and Computation: Practice and Experience, e7195, 2022.
- [8] **Cheng, X.**, Chen, Y., Cheng, B. X., et al. *An Emotion Cause Corpus for Chinese Microblogs with Multiple-user Structures*. ACM Transactions on Asian and Low-Resource Language Information Processing, 17(1):1–19, 2017.
- [9] Chen, Y., Cheng, B. X., **Cheng, X.** *Food Safety Document Classification Using LSTM-based Ensemble Learning*. Revista Tecnica de la Facultad de Ingenieria Universidad del Zulia, 39(10):172–178, 2016.

CONFERENCE PAPERS

- [1] **Cheng, X.**, Roy, U., Pandey A., Chakraborty N., Chakraborty S., & Calyam, P. *Heterogeneous Hypergraph Attention Network for Scholar Recommendation and Team Formation*. In *35th International ACM Conference on Knowledge and Information Management (CIKM 2026)*, pp. 1–11, 2026. **Acceptance Rate: 27%**
- [2] **Cheng, X.**, Edara, L. S., Zhang, Y., Kejriwal, M., & Calyam, P. *Influence Role Recognition and LLM-Based Scholar Recommendation in Academic Social Networks*. In *Proceedings of the 11th IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, pp. 1–11, 2024. **Acceptance Rate: 25%**
- [3] Shibo, B. W., Oruche, R., Goni, M. A., **Cheng, X.**, Milman, E., Calyam, P., & Kee, K. *Sifting Through Clutter and Finding Answers that Utter Among COVID-19 Data for Medical Users via Gateways*. 2023.
- [4] Neupane, K., Yeddulapalli, H. S., Kambhampati, A., **Cheng, X.**, Neupane, R. L., Kung, E. L., Rinehart, A., & Calyam, P. *Automation of News Content Curation and Storytelling for Local Newsrooms*. In *Proceedings of IEEE HPCC/DSS/SmartCity/DependSys*, pp. 1051–1060, 2023.
- [5] **Cheng, X.**, Zhang, Y., Joshi, H., Kejriwal, M., & Calyam, P. *Knowledge graph-based embedding for connecting scholars in academic social networks*. In *Proceedings of the 10th IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, pp. 1–10, 2023.
- [6] Wang, S., Neupane, R., Pandey, A., **Cheng, X.**, & Calyam, P. *Online Learning Platform for Application-Inspired Cloud and DevOps Curriculum*. In *Proceedings of IEEE HiPC Workshops*, pp. 35–42, 2021.
- [7] Oruche, R., Milman, E. D., **Cheng, X.**, Joish, M., Kulkarni, C., Sharma, A., Kee, K., Regunath, H., & Calyam, P. *Measurement of Utility in User Access of COVID-19 Literature via AI-powered Chatbot*. In *Proceedings of IEEE AIPR*, pp. 1–13, 2021.
- [8] Chen, Y., Hou, W., **Cheng, X.**, & Li, S. *Joint learning for emotion classification and emotion cause detection*. In *Proceedings of EMNLP*, pp. 646–651, 2018. **Acceptance Rate: 20.8%**
- [9] Chen, Y., Hou, W., & **Cheng, X.** *Hierarchical convolution neural network for emotion cause detection on microblogs*. In *ICANN 2018, Part I*, pp. 115–122. Springer, 2018.

ADVISING AND MENTORSHIP

- Upasana Roy (2025-2026), CS PhD Student at University of Missouri (Currently)
- Mahady Hasan Rayhan (2025), CS PhD Student at University of Missouri (Currently)
- Harshavardhan Chintapatla (2025-2026), CS Master Student at University of Missouri (Currently)
- Lakshimi Srinivas Edara (2023-2024), CS Master Student at University of Missouri, joined Spectrum
- Harsh Joshi (2022-2023), CS Master Student at University of Missouri, joined Cummins Inc.

PROFESSIONAL SERVICE

- Journal Peer-Reviewer: Concurrency and Computation: Practice and Experience 2025, 2026, Discover Artificial Intelligence (Springer Nature) 2025, 2026, IEEE Transactions on Network and Service Management (TNSM) 2023, 2024, 2025, Scientific Reports (Springer Nature) 2025.
- Conference Peer-Reviewer: IEEE International Conference on Network and Service Management (CNSM) 2026, IEEE International Conference on Computing, Networking and Communications (ICNC) 2024, 2025: Social Computing and Data Mining (SCDM), IEEE Conference on Network Function Virtualization and Software Defined Networks 2024, IEEE International Conference on Network Softwarization 2025, IEEE International Performance, Computing, and Communications Conference (IPCCC) 2024, IEEE International Conference on Communications (ICC) 2024, 2025.